

Green Bond Market Brief

What the labelled bond market actually is

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Data: Climate Bonds Initiative News Makers extract — 732 labelled issuances, 2015–2024, \$650.04bn. Every figure here is computed from the aggregate tables published alongside this brief, not quoted by hand; see Methodology.

Summary view

The labelled sustainable bond market is routinely presented as evidence that sustainable finance has reached scale. On the headline numbers it has: 732 transactions and \$650bn of issuance, with deal count growing from 10 in 2018 to 238 in 2023.

Disaggregated, the picture is narrower and more policy-dependent than the headline suggests:

- **82% of volume is sovereign.** Governments, not corporates, built this market. The private tail is \$113.8bn across 284 deals.
- **64% of volume is European.** Germany, France and the UK alone are 36.5%.
- **The instrument with the strongest theoretical claim is the smallest.** Sustainability-linked bonds are 15% of deals but **3.4% of volume**, averaging \$206m against green bonds' \$1,077m.
- **Average deal size has fallen 66%** since 2018, from \$1,757m to \$590m. The market is broadening — which is healthy — but broadening is not the private sector taking over.
- **The pricing advantage is real but tiny, and shrinking.** Measuring Germany's green twin Bunds over 8,094 paired daily observations gives a greenium of **–1.50bps**, down from –4.71bps in 2021 to under 1bp today.

The investment implication: labelled-debt supply is primarily a function of sovereign issuance programmes, not corporate ESG appetite. For anyone forecasting supply, the relevant calendar is the sovereign one.

1. Growth depends on what you count

The most-quoted growth figure for this market measures the wrong thing.

Measure	2018	2023	Growth
Deal count	10	238	+2,280%
USD volume	\$17.6bn	\$184.5bn	+950%
Average deal size	\$1,757m	\$775m	–56%

Deal count overstates growth by more than a factor of two, because average deal size fell as the market broadened. Both numbers are true; only the second describes how much capital moved.

This is not pedantry — count and volume rank the market differently. **China leads on deal count (68 transactions) but ranks sixth by volume**, because Chinese issuance in this dataset is many small bank deals while European volume sits in a handful of large sovereign programmes. An analysis built on counts would conclude the wrong thing about where this market is.

Annual volume, with 2024 as a partial year in this extract:

Year	Deals	Volume (\$bn)	Avg deal (\$m)
2018	10	17.6	1,757
2019	20	25.2	1,258
2020	36	44.9	1,247
2021	85	128.4	1,510
2022	169	140.3	830
2023	238	184.5	775
2024*	165	97.3	590

2. This is a sovereign market with a private tail

Sector	Deals	Volume (\$bn)	% of volume	Avg deal (\$m)
Sovereign	448	536.2	82.5%	1,197
Financials	95	46.2	7.1%	486
Unclassified	65	23.8	3.7%	366
Industrials	53	18.1	2.8%	341
All others	71	25.7	4.0%	362

Sovereign deals are **three times larger** on average than everything else (\$1,197m vs \$401m), so the sector's dominance of volume is a size effect as much as a frequency one.

Geographic concentration points the same way. Europe is 64.4% of volume across 293 deals; the top three countries are 36.5%:

Country	Deals	Volume (\$bn)	% of volume
Germany	58	95.2	14.6%
France	42	82.2	12.6%
UK	20	59.8	9.2%
Italy	16	45.9	7.1%
Chile	36	37.6	5.8%
China	68	33.1	5.1%

Read together with the sector table, the market's growth story is substantially the story of European sovereign green issuance programmes — the green Bund, the OAT verte, the green gilt — plus a cluster of emerging-market sovereigns (Chile, Mexico, Indonesia, Hungary) that have used the format for budget funding.

That matters for supply forecasting. Corporate ESG commitments are a weak predictor of labelled issuance; sovereign funding calendars are a strong one.

3. The SLB puzzle: the best instrument is the smallest

Sustainability-linked bonds have the stronger theoretical claim. Green bonds restrict *use of proceeds*, which says nothing about whether the issuer's overall emissions fall and requires a ring-fenceable project. SLBs tie the coupon to an *outcome* — a measured KPI — so in principle they work for any issuer and reward actual performance.

The market has not rewarded that logic with volume:

Theme	Deals	% of deals	Volume (\$bn)	% of volume	Avg deal (\$m)
Green	490	66.9%	527.7	81.2%	1,077
Sustainability	109	14.9%	72.5	11.2%	665
SLB	108	14.8%	22.2	3.4%	206
Social	25	3.4%	27.6	4.2%	1,105

SLBs match sustainability bonds on deal count but carry a fifth of the volume. The average SLB is **five times smaller** than the average green bond. SLB share of annual volume reached 11.8% in 2024 — real growth from nil before 2021, but from a very low base and in the partial year.

Why I think the incentive is too weak to scale. My companion analysis structures a sustainability-linked loan from first principles (same coupon-adjustment mechanism as an SLB). Three findings bear directly on this:

1. **The adjustment is small.** A ± 7.5 bps margin ratchet — mid-range for market practice of 2.5–15bps — sits inside the ordinary spread volatility of a BBB– credit. It is unlikely to be the binding factor in anyone's cost of capital. 2. **It scales with drawdown.** On a revolving facility the ratchet applies only to the drawn margin, so the benefit shrinks in direct proportion to how little is drawn. 3. **It is gated by a fixed cost.** Third-party KPI verification does not get cheaper when the saving falls. On

a £650m facility with ~£100k annual assurance cost, **break-even is 20.5% utilisation**: below that, the best possible sustainability outcome does not cover the cost of proving it.

If the pricing incentive is close to immaterial, the rational reasons to issue performance-linked debt are signalling, investor access and internal accountability. Those are real, but they are weaker drivers of volume than funding need — which is exactly what the 3.4% volume share looks like.

4. Greenium: what can and cannot be claimed

The academic literature consistently finds a small negative premium — green bonds yielding *less* than comparable conventional bonds:

Study	Market	Estimate (bps)	Method
Zerbib (2019)	Global, 110 bonds	–2	Matched-pair + 2-step OLS
Löffler et al. (2021)	Global primary	–15 to –20	Coarsened exact matching
Caramichael & Rapp (2022)	US corporates	–8	Matching + event study
Panizza et al. (2025)	Sovereign	–5 to –8	Synthetic control
Banque de France (2025)	Eurozone	–2 to –13	Propensity-score matching

Convention: (green yield – conventional yield); negative means a greenium exists.

These are not directly comparable — they differ in market, period, primary versus secondary pricing, and matching method. The spread of estimates is mostly heterogeneity, not disagreement about technique.

My own estimate: –1.50bps, and narrowing

I measured it rather than citing it. The method avoids the literature's central weakness: instead of modelling comparability, it uses Germany's green **twin** Bunds, where each green security has a conventional counterpart with the *same coupon, same maturity and same issuer*. The yield difference is then the greenium almost by construction, with no matching model doing the work.

All nine green Federal securities outstanding were matched to their exact twins — coupon and maturity identical on both legs — giving **8,094 paired daily observations from September 2020 to October 2026**, all from the issuer's own published data.

Pooled greenium	–1.50bps
HAC standard error	0.052
t-statistic	–28.7
Paired observations	8,094 across 9 twin pairs
Days with a negative spread	99.8%
Range of per-pair means	–0.65bps to –2.40bps

The greenium is small, highly persistent, and statistically unambiguous. It shows no maturity pattern (correlation between years-to-maturity and mean greenium: 0.11), so it is a label effect rather than a term-structure artefact.

Inference matters here. A daily yield spread is strongly autocorrelated. The ordinary standard error gives $t = -94$; the Newey-West HAC standard error is **3.3× larger** and gives $t = -28.7$. The conclusion survives either way, but an uncorrected standard error would have overstated the precision threefold — and on a shorter sample it could easily manufacture significance that was not there.

The more interesting result is the trend. The greenium has compressed by roughly 80% from its 2021 peak:

Year	Mean greenium	Obs
2020	–2.35bps	80
2021	–4.71bps	495
2022	–1.94bps	855
2023	–2.27bps	1,331
2024	–1.07bps	1,695

Year	Mean greenium	Obs
2025	−0.76bps	1,955
2026	−0.95bps	1,683

This is consistent with a maturing market: as green supply grew and the novelty premium faded, the pricing advantage eroded. It also sharpens the argument above. At −4.7bps there was at least a case that the label carried a funding benefit. At under 1bp, the benefit is inside the bid-offer spread on most days — too small to be a reason to issue, which is consistent with a market whose growth is driven by sovereign funding strategy rather than by price.

My estimate sits close to Zerbib's −2bps and well below the primary-market studies' −15 to −20bps. That gap is expected rather than contradictory: those studies measure the concession at issuance, while this measures secondary-market trading in the most liquid sovereign curve in Europe, where any mispricing is arbitrated hardest.

5. What would change my view

The honest limitations of this analysis:

- **The dataset is a curated newsflow extract**, not an exhaustive market census. The Climate Bonds Initiative News Makers series tracks notable transactions, and large sovereign deals are more likely to be *notable*. The 82% sovereign share may therefore overstate the true market share — this is the single largest threat to the conclusion, and it cannot be resolved from this dataset alone. Reconciling against a full issuance database is the first thing I would do next.
- **2024 is partial**, so the recent trend and the 11.8% SLB share are provisional.
- **Sector is undisclosed for 65 deals** (\$23.8bn, 3.7% of volume), shown as Unclassified rather than dropped or allocated.
- **Two source records are known to be wrong** and are flagged rather than silently corrected: one issuer is tagged to the wrong country, and one 2015 transaction is labelled SLB years before that market existed.
- **The greenium estimate is German sovereign only**. It is the cleanest available estimate — the twin structure removes the matching problem rather than modelling around it — but it does not generalise to corporate issuers, to other sovereigns, or to primary-market pricing, where the published estimates are several times larger.

Evidence that would change the conclusion: a full-market database showing corporate issuance materially larger than this extract implies; or SLB volume share continuing above ~10% through complete years, which would suggest the incentive problem matters less than I argue.

Methodology

Data was cleaned and aggregated in Python (pandas) and the model built with openpyxl; every figure in this brief regenerates from the published aggregate tables with one command, and the aggregations are covered by 334 unit tests. Each figure quoted here is asserted against those tables in the test suite, so this document cannot drift from the data behind it.

The source extract required substantive normalisation before it could be aggregated, all documented on the model's Data Quality sheet: sector "not disclosed" was encoded as the string "0" (64 records); 34 sector labels denoted 11 real groups (seven spellings of the financial sector each ranked separately); and four country labels were duplicated — USA/United States, UK/United Kingdom, China_HK, and a misspelt Supranational, plus one trailing-space variant of Netherlands. Left uncorrected these understated the UK by \$0.45bn and inflated the country count from 64 to an apparent 69.

The greenium estimate uses the Deutsche Finanzagentur's published daily price and yield series for each green Federal security and its conventional twin; the twin pairing is derived by matching coupon and maturity exactly, not asserted. `scripts/fetch_bund_yields.py` reproduces the download and `scripts/greenium.py` the estimate.

Sources are public throughout: Climate Bonds Initiative for issuance, published academic papers for the comparative greenium evidence, and the Deutsche Finanzagentur for bond yields. No proprietary terminal data is used, and the repository's CI fails the build if a reference to one appears — including inside the Excel files.

Public access is not the same as permission to republish, and two of these sources reserve their rights: CBI prohibits reproducing their content without written permission, and Finanzagentur marks its published data all rights reserved. Neither dataset is redistributed in the repository. What is published instead is this project's own derived output — the aggregate tables and the greenium estimator results — which is what every figure above is computed from and checked against. Both are regenerable from their publishers in one command by anyone with access. The repository's DATA.md records the terms and

the trade-off.

Repository: github.com/ShangyuZ/Sustainable-finance-portfolio

References

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